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Basic Mathematics
BCS-12

Question: For what value of 'k' the points $(-k+1, 2k)$, $(k, 2-2k)$ and $(-4-k, 6-2k)$ are collinear.

Answer:

$$\text{Area of triangle } \Delta = \frac{1}{2}\left|\begin{vmatrix} -k+1 & 2k & 1 \\ k & 2-2k & 1 \\ -4-k & 6-2k & 1 \end{vmatrix}\right|$$

$$= \frac{1}{2}\begin{vmatrix} -k+1 & 2k & 1 \\ 2k-1 & 2-4k & 0 \\ -5 & 6-4k & 0 \end{vmatrix}$$

$$= \frac{1}{2}\left|\begin{vmatrix} 2k-1 & 2-4k \\ -5 & 6-4k \end{vmatrix}\right|$$

$$= \frac{1}{2}(2k-1)(6-4k) + 5(2-4k)$$

$$= \frac{1}{2}\left|-8k^2 - 4k + 4\right| = \left|4k^2 + 2k - 2\right|$$

These points are collinear if $\Delta = 0$

i.e., if $\left|4k^2 + 2k - 2\right| = 0$

i.e., if $2(2k-1)(k+1) = 0$

i.e. if $k = -1, \frac{1}{2}$

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2 Question :- Solve the following system of equations by using Matrix Inverse Method.

$$3x+4y+7z=14$$
$$2x-y+3z=4$$
$$2x+2y-3z=0$$

Answer :- Here $3x+4y+7z=14$
$2x-y+3z=4$
$2x+2y-3z=0$

Now converting given equations into matrix form

$$\begin{bmatrix} 3 & 4 & 7 \\ 2 & -1 & 3 \\ 2 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 14 \\ 4 \\ 0 \end{bmatrix}$$

Now, $A = \begin{bmatrix} 3 & 4 & 7 \\ 2 & -1 & 3 \\ 2 & 2 & -3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 14 \\ 4 \\ 0 \end{bmatrix}$

$\therefore AX = B$

$\therefore X = A^{-1}B$

$$|A| = \begin{vmatrix} 3 & 4 & 7 \\ 2 & -1 & 3 \\ 2 & 2 & -3 \end{vmatrix}$$

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$$= 3\times\begin{vmatrix}-1 & 3\\ 2 & -3\end{vmatrix} - 4\times\begin{vmatrix}2 & 3\\ 2 & -3\end{vmatrix} + 7\times\begin{vmatrix}2 & -1\\ 2 & 2\end{vmatrix}$$

$$= 3\times(-1\times(-3)-3\times2) - 4\times(2\times(-3)-3\times2) + 7\times(2\times2-(-1)\times2)$$

$$= 3\times(3-6) - 4\times(-6-6) + 7\times(4+2)$$
$$= 3\times(-3) - 4\times(-12) + 7\times(6)$$
$$= -9+48+42$$
$$= 81$$

Here, $|A| = 81 \neq 0$

$\therefore$ A^{-1} is possible

$$Adj(A) = Adj\begin{bmatrix}3 & 4 & 7\\ 2 & -1 & 3\\ 2 & 2 & -3\end{bmatrix}$$

$$= \begin{bmatrix} +\begin{vmatrix}1 & 3\\ 2 & -3\end{vmatrix} & -\begin{vmatrix}2 & 3\\ 2 & -3\end{vmatrix} & +\begin{vmatrix}2 & -1\\ 2 & 2\end{vmatrix} \\ -\begin{vmatrix}4 & 7\\ 2 & -3\end{vmatrix} & +\begin{vmatrix}3 & 7\\ 2 & -3\end{vmatrix} & -\begin{vmatrix}3 & 4\\ 2 & 2\end{vmatrix} \\ +\begin{vmatrix}4 & 7\\ -1 & 3\end{vmatrix} & -\begin{vmatrix}3 & 7\\ 2 & 3\end{vmatrix} & +\begin{vmatrix}3 & 4\\ 2 & -1\end{vmatrix}\end{bmatrix}^T$$

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$$= \begin{bmatrix} +(-1\times(-3)-3\times2) & -(2\times(-3)-3\times2) & +(2\times2-(-1)\times2) \\ -(4\times(-3)-7\times2) & +(3\times(-3)-7\times2) & -(3\times2-4\times2) \\ +(4\times3-7\times(-1)) & -(3\times3-7\times2) & +(3\times(-1)-4\times2) \end{bmatrix}^T$$

$$= \begin{bmatrix} +(3-6) & -(-6-6) & +(4+2) \\ -(-12-14) & +(-9-14) & -(6-8) \\ +(12+7) & -(9-14) & +(-3-8) \end{bmatrix}^T$$

$$\begin{bmatrix} -3 & 12 & 6 \\ 26 & -23 & 2 \\ 19 & 5 & -11 \end{bmatrix}^T$$

Now, $A^{-1} = \frac{1}{|A|} \times Adj(A)$

Here, $X = A^{-1} \times B$

$\therefore X = \frac{1}{|A|} \times Adj(A) \times B$

$$= \frac{1}{81} \times \begin{bmatrix} -3 & 26 & 19 \\ 12 & -23 & 5 \\ 6 & 2 & -11 \end{bmatrix} \times \begin{bmatrix} 14 \\ 4 \\ 0 \end{bmatrix}$$

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$$= 0.0123 \times \begin{bmatrix} -3\times14 + 26\times4 + 19\times0 \\ 12\times14 - 23\times4 + 5\times0 \\ 6\times14 + 2\times4 - 11\times0 \end{bmatrix}$$

$$= 0.0123 \times \begin{bmatrix} 62 \\ 76 \\ 92 \end{bmatrix}$$

$$= \begin{bmatrix} 0.7654 \\ 0.9383 \\ 1.1358 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0.7654 \\ 0.9383 \\ 1.1358 \end{bmatrix}$$

$$\therefore x = \frac{62}{81},\ y = \frac{76}{81},\ z = \frac{92}{81}$$

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Q3 Question: Use principle of Mathematical Induction to prove that:

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1}$$

Answer: Let P_n denote the statement

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\cdots\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1}$$

When $n=1$, P_n becomes $\frac{1}{1\cdot 2}=\frac{1}{1+1}$ or $\frac{1}{2}=\frac{1}{2}$

This shows that the result holds for $n=1$. Assume that P_k is true for some $k\in N$. That is assume that

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\cdots\cdots+\frac{1}{k(k+1)}=\frac{k}{(k+1)} \quad -(1)$$

We shall now show that ~~that~~ the true of P_k implies the truth of P_{k+1} where P_{k+1} is

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\cdots+\frac{1}{k(k+1)}+\frac{1}{(k+1)(k+2)}=\frac{k+1}{k+2}$$

$$\text{LHS of (1)}=\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\cdots\frac{1}{k(k+1)}+\frac{1}{(k+1)(k+2)}$$

$$=\frac{k}{(k+1)}+\frac{1}{(k+1)(k+2)} \quad \left[\begin{array}{c}\text{induction}\\ \text{assumption}\end{array}\right]$$

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$$= \frac{k(k+2)+1}{(k+1)(k+2)} = \frac{k^2+2k+1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

= RHS of (1)

This shows that the result holds for $n = k+1$; therefore, the truth of P_k implies the truth of P_{k+1}. The two steps required for a proof by mathematical induction have been completed, so our statement is true for each natural number n.

4 Question:- How many terms of G.P $\sqrt{3}, 3, 3\sqrt{3}, ---$ --- Add upto $39+13$

Answer:- Here $a = \sqrt{3}$, $ar = 3$, so that $r = \sqrt{3}$.

Let $39 + 13\sqrt{3}$ be the sum to n terms of the given G.P, that is,

$$S_n = 39 + 13\sqrt{3}$$

$$\Rightarrow \frac{a(r^n - 1)}{r-1} = 39 + 13\sqrt{3} \Rightarrow \frac{(\sqrt{3})\left[(\sqrt{3})^n - 1\right]}{\sqrt{3}-1}$$

$$\Rightarrow 13\sqrt{3}(\sqrt{3}+1)$$

$$\Rightarrow 3^{n/2} = 1 + 13(3-1) = 1 + 26 = 27 = 3^3$$

$$\Rightarrow \frac{n}{2} = 3 \text{ or } n = 6.$$

Thus, 6 terms of $\sqrt{3}, 3, 3\sqrt{3}$ ------ are

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required to obtain a sum of $39+13\sqrt{13}$.

5 Question: If $y = ae^{mx} + be^{-mx}$, Prove that $\frac{d^2y}{dx^2} = m^2y$

Answer:- We have $y = ae^{mx} + be^{-mx}$

Differentiating both sides with respect to x, we get $\frac{dy}{dx} = \frac{d}{dx}(ae^{mx} + be^{-mx}) =$
$ame^{mx} - bme^{-mx}$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dx}(ame^{mx} - bme^{-mx})$$

$$= am^2e^{mx} - bm(-m)e^{-mx}$$

$$\Rightarrow am^2e^{mx} + bm^2e^{-mx}$$

$$\Rightarrow m^2(ae^{mx} + be^{-mx})$$

$$\Rightarrow m^2y$$

6 Question:- Integrate function $f(x) = \frac{x}{[(x+1)(2x-1)]}$ w.r.t. x

Answer: We first resolve the integrand into partial fractions. Write

$$\frac{x}{(x+1)(2x-1)} = \frac{A}{(x+1)} + \frac{B}{(2x-1)}$$

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$$\Rightarrow x = A(2x-1) + B(x+1)$$

Put $x = \frac{1}{2}$ and -1 to obtain

$$\frac{1}{2} = B\left(\frac{1}{2}+1\right) \Rightarrow B = \frac{1}{3}$$

$$-1 = A(-3) \Rightarrow A = \frac{1}{3}$$

Thus,

$$\int \frac{x}{(x+1)(2x-1)}dx = \frac{1}{3}\int \frac{dx}{(x+1)} + \frac{1}{3}\int \frac{dx}{(2x-1)}$$

$$= \frac{1}{3}\log|x+1| + \frac{1}{3}\cdot\frac{1}{2}\log|2x-1| + C$$

$$= \frac{1}{3}\log|x+1| + \frac{1}{6}\log|2x-1| + C$$

7. Question: If $1, \omega, \omega^2$ are cube Roots of unity show that $(1+\omega)^2 - (1+\omega)^3 + \omega^2 = 0$.

Answer: As $1+\omega+\omega^2 = 0$, we get

$$1+\omega = -\omega^2 \text{ and } 1+\omega^2 = -\omega$$

Thus,

$$(1+\omega)^2 - (1+\omega^2)^3 + \omega^2$$

$$= (-\omega^2)^2 - (-\omega)^3 + \omega^2$$

$$= \omega^4 + \omega^3 + \omega^2 = \omega^3\omega + 1 + \omega^2$$

$$= \omega + 1 + \omega^2 = 0$$

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8 Question: If α, β are roots of equation $2x^2-3x-5=0$, then find Quadratic equation whose roots are α^2, β^2

Answer:- Since α and β are roots of $2x^2-3x-5=0$

$$\alpha+\beta = \frac{3}{2} \text{ and } \alpha\beta = \frac{-5}{2}$$

We are to form a quadratic equation whose roots are α^2, β^2.

Let S = Sum of roots $= \alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$.

$$= \left(\frac{3}{2}\right)^2 - 2\left(-\frac{5}{2}\right) = \frac{9}{4}+5 = \frac{29}{4}$$

P = Product of roots $= \alpha^2\beta^2 = (\alpha\beta)^2 = \left(\frac{-5}{2}\right)^2 = \frac{25}{4}$.

Putting values of S and P in $x^2 - Sx + P = 0$, the required equation is

$$x^2 - \left(\frac{29}{4}\right)x + \frac{25}{4} = 0 \text{ or } 4x^2 - 29x + 25 = 0.$$

9 Question: Solve the inequality $\frac{3}{5}(x-2) \leq \frac{5}{3}(2-x)$ and graph the solution set.

Answer: Given:

$$\frac{3}{5}(x-2) \leq \frac{5}{3}(2-x)$$

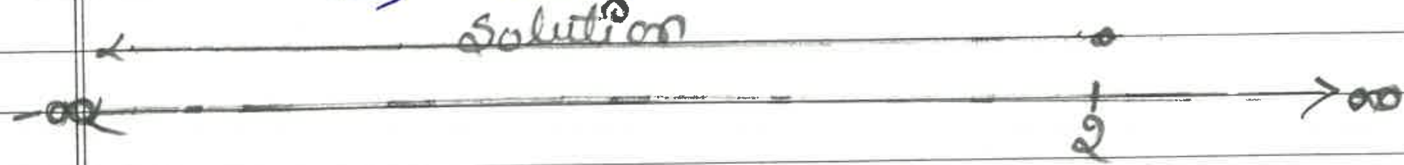
$$\Longrightarrow 9(x-2) \leq 25(2-x)$$

$\Rightarrow 9x - 18 < 50 - 25x$

$\Rightarrow 25x + 9x < 50 + 18$

$\Rightarrow 34x < 68$

$\Rightarrow x < 2$



$\therefore$ Solution set, $x \in (-\infty, 2$

10 Question: If a positive number exceeds its positive square root by 12, then find the number.

Answer:- Let the positive number be x. According to the given condition

$$x - \sqrt{x} = 12 \Rightarrow x - 12 = \sqrt{x}$$

Squaring both the sides, we $x^2 - 24x + 144 = x$

$\Rightarrow x^2 - 25x + 144 = 0 \Rightarrow x^2 - 9x - 16x + 144 = 0$

$\Rightarrow x(x-9) - 16(x-9) = 0 \Rightarrow (x-16)(x-9) = 0$

$\Rightarrow x = 16$ or 9.

Putting $x = 16$ in (1), we get $16 - 4 = 12$ which is true.
Thus, $x = 16$.

11 Question: Find the area bounded by the curves $x^2 = y$ and $y = x$.

Answer:- To obtain point of intersection of $y = x^2$

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and $y = x$, we set

$$x^2 = x \Rightarrow x(x-1) = 0 \Rightarrow x = 0, 1$$

Thus, the two curves intersect in $(0,0)$ and $(1,1)$. See fig

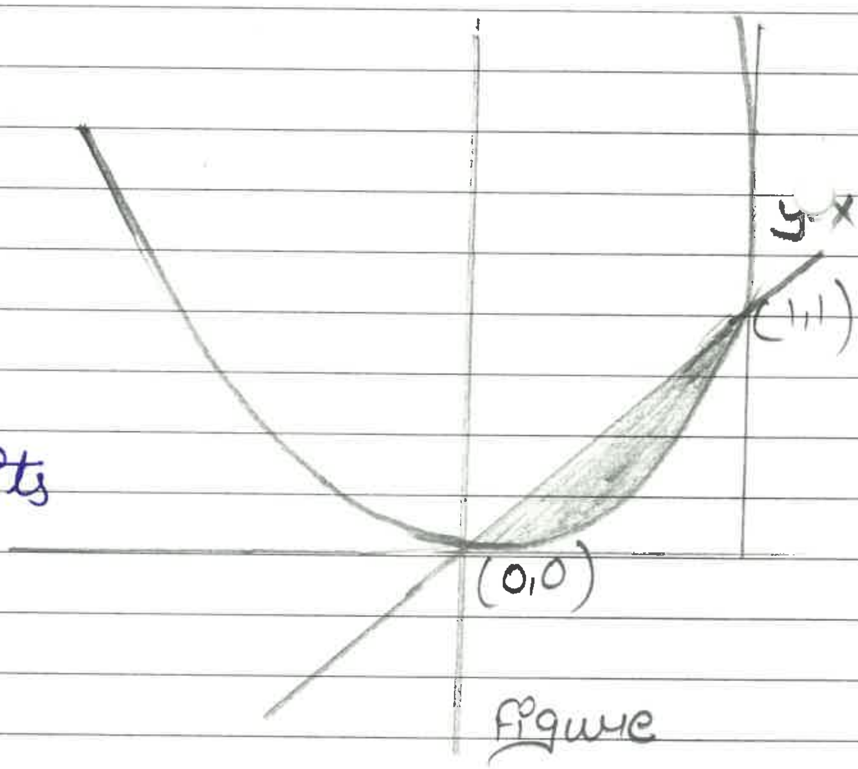
Required area

$$= \int_0^1 (x - x^2)\,dx$$

$$= \left[\frac{1}{2}x^2 - \frac{1}{3}x^3\right]_0^1$$

$$= \left(\frac{1}{2} - \frac{1}{3}\right) \text{ sq. units}$$

$$= \frac{1}{6} \text{ sq. units.}$$



figure

12 Question: Find the inverse of the matrix $A =$

$$\begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix},$$ if it exists.

Answer: Consider the identity $A = I_3 A$

$$\begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

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Again, applying $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 + R_1$, we get

$$\begin{bmatrix} 1 & 6 & 4 \\ 0 & -8 & -9 \\ 0 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} A$$

Again, applying $R_3 \to R_3 + R_2$, we have

$$\begin{bmatrix} 1 & 6 & 4 \\ 0 & -8 & -9 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} A$$

Since, we have obtain a row of zeros on the L.H.S., we see that A cannot be reduced to an identity matrix. Thus, A is not invertible, infact, note that A is a singular matrix as $|A| = 0$.

13 Question: If m times the m^{th} term of an A.P. is n times its n^{th} term, show that $(m+n)^{th}$ term of the A.P. is zero.

Answer: We are given that $ma_m = na_n$

$$\Rightarrow m[a + (m-1)d] = n[a + (n-1)d]$$

$$\Rightarrow m[a + md - d] = n[a + nd - d]$$

$$\Rightarrow m[a + md] = n[a + nd] \text{ where } a = a - d$$

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$\Rightarrow ma + m^2 d = \cancel{m}\,(na + n^2 d \Rightarrow (m-n)a + (m^2-n^2)d = 0$

$\Rightarrow (m-n)[a + (m+n)d] = 0$

$\Rightarrow a + (m+n)d = 0$

$\Rightarrow a + (m+n-1)d = 0 \quad [\because a = a-d]$

left hand side is nothing but the $(m+n)$th term of the A.P.

14 Question: Show that

(i) $\lim_{n \to 0} \frac{|x|}{x}$ does not exist

Answer: Let $f(x) = \frac{|x|}{x}, \; x \neq 0$

Since $|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$

$\therefore f(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$

So, $\lim_{x \to 0^+} f(x) = \lim_{x \to 0} (1) = 1$ and

$\lim_{x \to 0^-} f(x) = \lim_{x \to 0} (-1) = -1$

Thus $\lim_{x \to 0} f(x)$ does not exist.

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ii) $f(x) = |x|$ is continuous at $x = 0$.

Answer:- Recall that

$$f(x) = |x| \begin{cases} x, & x \geq 0 \\ -x & x < 0 \end{cases}$$

To show that f is continuous at $x=0$, it is sufficient to show that

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+}(x) = f(0) \text{ and}$$

We have

$$\lim_{x \to 0^-} f(x) = \lim_{h \to 0^+} f(0-h) = \lim_{h \to 0^+} f(-h)$$

$$= \lim_{h \to 0^+} -(-h)$$

$$= \lim_{h \to 0^+} h = 0$$

$$\text{and } \lim_{x \to 0^+} f(x) = \lim_{h \to 0^+} f(0+h) = \lim_{h \to 0^+} f(h)$$

$$= \lim_{h \to 0^+} (h) = 0.$$

Thus, $\lim_{h \to 0^-} f(x) = \lim_{h \to 0^+} f(x) = 0$

Also, $f(0) = 0$

Therefore, $\lim_{x \to 0^+} f(x) = 0 = f(0)$

Hence, f is continuous at $x = 0$.

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15 Question:- Suriti wants to Invest at most 12000 in saving certificates and National Saving Bonds. She has to invest at least 2000 in Saving certificates and at least 4000 in National Saving Bonds. If Rate of Interest on saving certificates in 8% per annum and rate on national saving bond is 10% per annum. How much money should she invest to earn maximum yearly income? Find also the maximum yearly income.

Answer:- Let Suriti

invests Rs x in Saving certificates and Rs y in National Saving

Therefore,

$x, y \geq 0$

Suriti wants to invest at most Rs 12000 in Saving certificates and National

$$x + y \leq 12000$$

According to rules, he has to invest at least Rs 2000 in Saving certificates and at least Rs. 4000 in National Saving Bonds.

$$x \geq 2000$$

$$y \geq 4000$$

If the rate of interest on saving certificate

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is 8% per annum and the rate of interest on National Saving Bond is 10% per annum.

Total earning from investment $= Z = \frac{8x}{100} + \frac{10y}{100}$

which is to be maximised.

Thus, the mathematical formulation of the given linear programming problem is

$$\text{Max } Z = \frac{8x}{100} + \frac{10y}{100}$$

subject to

$$x + y \leq 12000$$
$$x \geq 2000$$
$$y \geq 4000$$
$$x, y \geq 0$$

First we will convert inequation into equations as follows:

$x + y = 12000$, $x = 2000$, $y = 4000$, $x = 0$ and $y = 0$

Region represented by $x + y \leq 12000$:

The line $x + y = 12000$ meets the coordinates axes at $A(12000, 0)$ and $B(0, 12000)$ respectively. By joining these points we obtain the line $x + y = 12000$. Clearly $(0,0)$ satisfies the inequation $x + y \leq 12000$. So, the region which contains the origin represents the solution set of the inequation $x+y \le 12000$.

Region represented by $x \ge 2000$:

The line $x = 2000$ is the line that passes through $(2000, 0)$ and is parallel to y-axis. The region to the right of the line $x = 2000$ will satisfy the inequation $x \ge 2000$.

Region represented by $y \ge 4000$:

The line $y = 4000$ is the line that passes through $(0, 4000)$ and is parallel to x axis. The region above the line $y = 4000$ will satisfy the inequation $y \ge 4000$.

Region represented by $x \ge 0$ and $y \ge 0$:

Since, every point in the first quadrant satisfies that inequations. So, the first quadrant is the region represented by the inequations $x \ge 0$, and $y \ge 0$.

The feasible region determined by the system of constraints is

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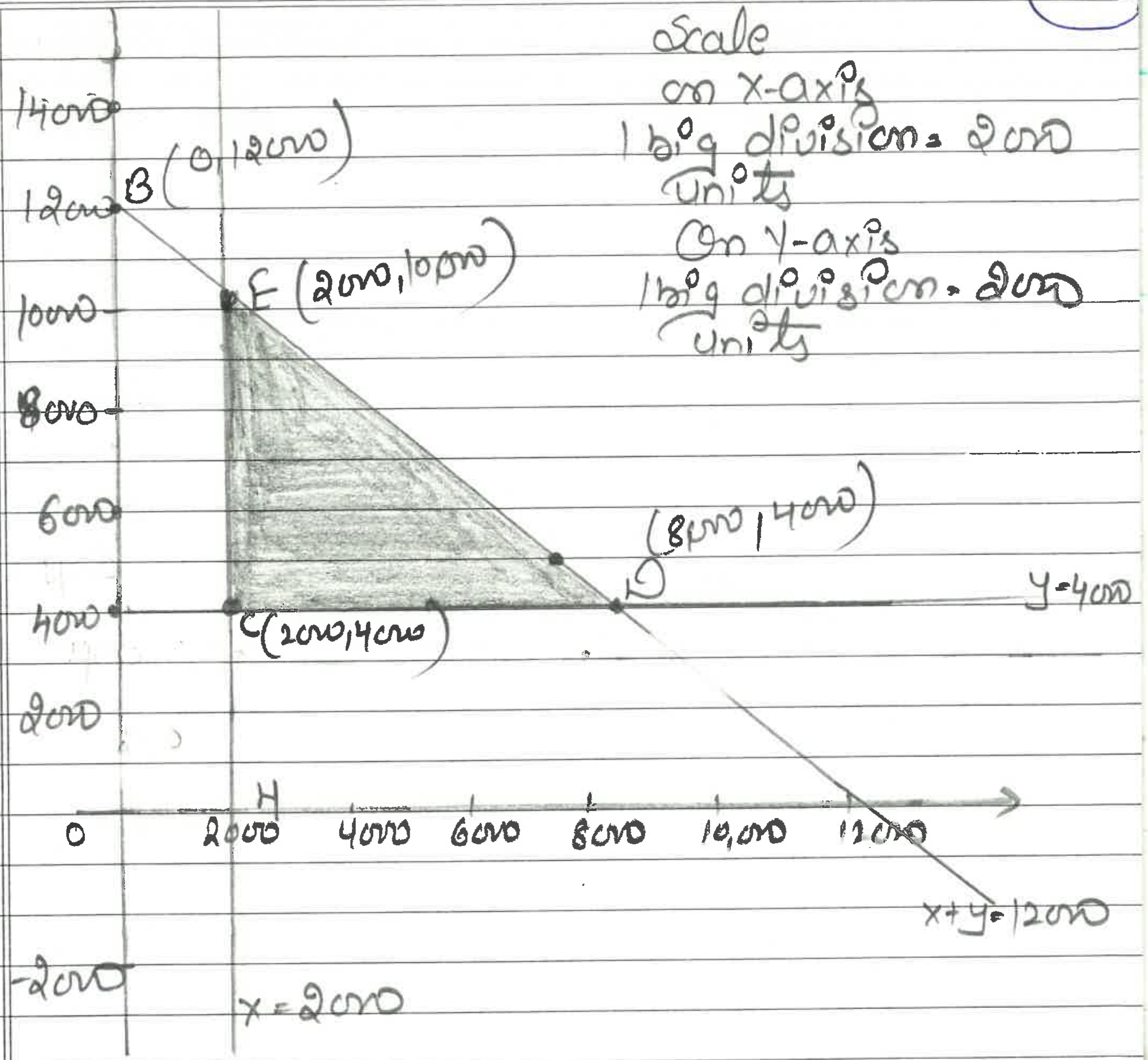
Scale
on x-axis
1 big division = 2000 units
On Y-axis
1 big division = 2000 units



The corner points are E(2000, 10000), D (8000,4000), C(2000,4000)

The values of Z at these corner points are as follows

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Corner point	$Z = \frac{8x}{100} + \frac{10y}{100}$
E	1160
D	1040
C	560

The maximum value of Z is 1160 which is attained at E (2000, 10,000). Thus, the maximum earning is Rs 1160 obtained when Rs 2000 were invested in Saving's certificates and Rs 10000 were invested in National Saving Bond.

16 Question:- A spherical ballon is being inflated at the rate of 900 cm3/sec. How fast is the Radius of the ballon increasing when the Radius is 15 cm.

Answer:- We are given the rate of change of volume and are asked the rate of change of the radius. Let r cm be the radius and V cubic centimeters be the volume of the ballon at instant t.

Thus, $V = \frac{4}{3}\pi r^3$

Differentiating both the sides of ~~the~~ this equation with respect to t, we get

$$\frac{dV}{dt} = \frac{4}{3}\pi(3r^2)\frac{dr}{dt} = 4\pi r^2\frac{dr}{dt}$$

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Note that have used the chain rule on the right hand side.

It is given that $\frac{dv}{dt} = 900$ and $r = 15$. Thus,

$$900 = 4\pi(15)^2 \frac{dr}{dt}$$

$$\rightarrow \frac{dr}{dt} = \frac{900}{4\pi(225)} = \frac{1}{\pi} \text{ cm/sec.}$$

Thus, the radius is increasing at the rate of $= \frac{1}{\pi}$ cm/sec.

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